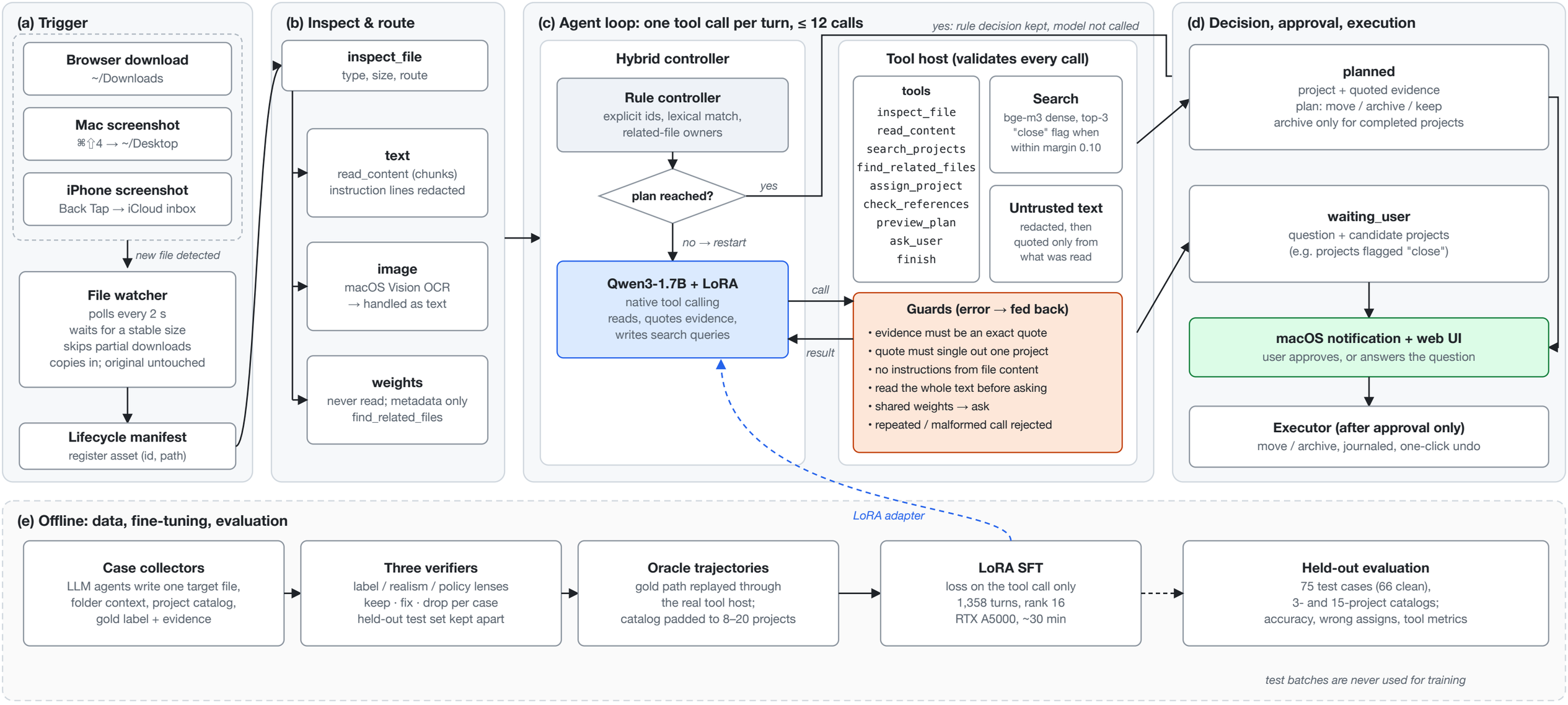

(a) Trigger
Browser download
~/Downloads
Mac screenshot
⌘⇧4 → ~/Desktop
iPhone screenshot
Back Tap → iCloud inbox
new file detected
File watcher
polls every 2 s
waits for a stable size
skips partial downloads
copies in; original untouched
Lifecycle manifest
register asset (id, path)
(b) Inspect & route
inspect_file
type, size, route
text
read_content (chunks)
instruction lines redacted
image
macOS Vision OCR
→ handled as text
weights
never read; metadata only
find_related_files
(c) Agent loop: one tool call per turn, ≤ 12 calls
yes: rule decision kept, model not called
Hybrid controller
Rule controller
explicit ids, lexical match,
related-file owners
plan reached?
yes
no → restart
Qwen3-1.7B + LoRA
native tool calling
reads, quotes evidence,
writes search queries
call
result
Tool host (validates every call)
tools
inspect_file
read_content
search_projects
find_related_files
assign_project
check_references
preview_plan
ask_user
finish
Search
bge-m3 dense, top-3
"close" flag when
within margin 0.10
Untrusted text
redacted, then
quoted only from
what was read
Guards (error → fed back)
• evidence must be an exact quote
• quote must single out one project
• no instructions from file content
• read the whole text before asking
• shared weights → ask
• repeated / malformed call rejected
(d) Decision, approval, execution
planned
project + quoted evidence
plan: move / archive / keep
archive only for completed projects
waiting_user
question + candidate projects
(e.g. projects flagged "close")
macOS notification + web UI
user approves, or answers the question
Executor (after approval only)
move / archive, journaled, one-click undo
(e) Offline: data, fine-tuning, evaluation
LoRA adapter
Case collectors
LLM agents write one target file,
folder context, project catalog,
gold label + evidence
Three verifiers
label / realism / policy lenses
keep · fix · drop per case
held-out test set kept apart
Oracle trajectories
gold path replayed through
the real tool host;
catalog padded to 8–20 projects
LoRA SFT
loss on the tool call only
1,358 turns, rank 16
RTX A5000, ~30 min
Held-out evaluation
75 test cases (66 clean),
3- and 15-project catalogs;
accuracy, wrong assigns, tool metrics
test batches are never used for training